

TrainerTap: Weightlifting Support System Prototype Simulating Personal Trainer's Tactile and Auditory Guidance

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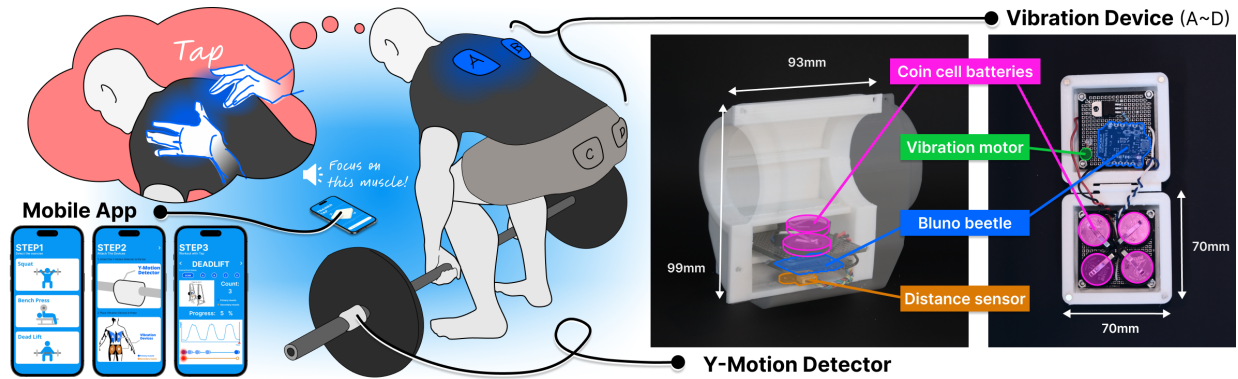


Figure 1: *TrainerTap* is a weightlifting support system that facilitates solo training by replicating the trainer's tapping action through vibration devices in pockets while providing auditory guidance via a mobile app based on the user's vertical motion detected using a y-motion detector.

ABSTRACT

Working out alone at the gym often lacks the quality and intensity of exercises compared to the training session with a personal trainer. To narrow this gap, we introduce *TrainerTap*, which simulates the personal trainer's presence during solitary weightlifting workouts. *TrainerTap* replicates the trainer's manual interventions of tapping the trainee's body parts to capture their attention on target muscles and provides auditory guidance to support executing the movements at a consistent tempo.

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UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

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ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3616644>

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

weightlifting, haptic wearable, mind-muscle connection

ACM Reference Format:

Hye-Young Jo, Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son, and Yoonji Kim. 2023. *TrainerTap*: Weightlifting Support System Prototype Simulating Personal Trainer's Tactile and Auditory Guidance. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3616644>

1 INTRODUCTION

Unlike trainer-led workouts that ensure effective muscle gain and sustained motivation [5, 7], self-guided weightlifting often falls short in terms of exercise quality and intensity [2]. To improve the

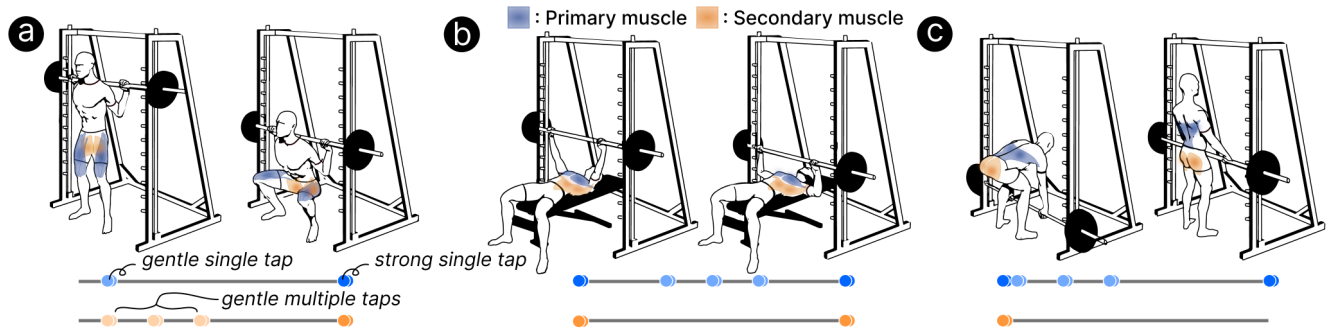


Figure 2: *TrainerTap*'s tapping pattern designs of the big three workouts: (a) squat (b) bench press (c) deadlift

solitary workout experience, prior works focused on self-error identification by comparing the user motion with the expert's [1, 3], visualizing the muscle activity [4, 10], and projecting the user motion in the environment [9]. In contrast, we focus on physical interaction. During sessions, trainers use their hands to touch the trainees' body parts, encouraging them to consciously engage in muscle movements through the mind-muscle connection [6, 8]. Alongside these tactile cues, trainers also provide auditory encouragement, such as counting, to pace and control the intensity of exercises. We take the approach of enhancing this sensory experience.

TrainerTap is a weightlifting support system that replicates the personal trainer's tapping actions and voice commands. Our work enhances solitary training by creating an illusion of a virtual trainer guiding when and where to engage muscles. We explore the vibration modality beyond error notification, using it as a physical language to capture the personal trainer's touch techniques. We believe that *TrainerTap* has the potential to support both solitary and trainer-led workouts by promoting effective muscle usage while respecting personal space.

2 THE TRAINERTAP SYSTEM

The *TrainerTap* system consists of (1) a *y-motion detector* for sensing user motion, (2) *vibration devices* for simulating the trainer's tactile feedback, and (3) a *mobile app* for facilitating communication between devices, providing auditory feedback, and offering a user interface (Figure 1). Our prototype focuses on providing tactile guidance on two muscle groups—the primary and secondary muscle groups (left/right)—in the big three workouts (squat, bench press, deadlift) on the Smith machine.

Walkthrough: First, the user wears customized gym clothes with eight pockets and selects their desired exercises in the mobile app. Following the on-screen instruction, the user places the vibration devices inside the designated pockets and attaches the *y-motion detector* to the barbell. For the first three repetitions, the *y-motion detector* calibrates the bar height to determine the user's range of motion. Then it tracks the user's progress and counts the repetition. The mobile app receives this motion data and sends signals to the vibration devices to generate the tapping stimulus accordingly. In addition, it plays auditory guidance, such as counting the movement, breathing instruction, and cheering the user.

Implementation: We used the Bluno Beetle, a compact Arduino (33.2mm x 28.9mm) with Bluetooth capability, and CR2032 coin-type lithium batteries for all devices. Additionally, the *y-motion detector* has a laser range distance sensor (TOF10120), and the vibration devices include a coin-type vibration motor (BL-0825VP).

All devices connect to the mobile app (JAVA) via Bluetooth. The *y-motion detector* receives data from the distance sensor and transmits it through the app to multiple vibration devices. The vibration devices simulate tap patterns using the vibration motor based on the received motion type and Y-motion data.

3 TAPPING PATTERN DESIGN

We designed tactile instructions for the *TrainerTap* with three trainers with 5-8 years ($M : 6.33, SD : 1.53$) of teaching experience.

TrainerTap has three tapping patterns for primary and secondary muscles in three popular workouts (squat, bench press, and deadlift) that capture different intentions from the trainers as follows:

- A **strong single tap** means to exert the maximum force of the target muscle (vibration 15000 rpm).
- A **gentle single tap** means to be aware of the muscle and maintain its tension (vibration 8800 rpm).
- A **gentle multiple taps** shares the same intention as a *gentle single tap* but includes multiple stimuli to raise awareness of less frequently used muscles (e.g., back).

Squat with *TrainerTap*

The squat targets the quadriceps (primary muscles) and the gluteal muscles (secondary muscles). As shown in Figure 2(a), the system delivers gentle taps to both the primary and secondary muscles at the beginning to remind users to consciously maintain tension in both muscle groups. Then, additional taps are given for the secondary muscles to engage the buttock area, which is commonly underdeveloped compared to the thigh muscles. When the user reaches the lowest point, the system delivers a strong tap to both muscles, prompting the user to stand up and exert maximum power.

Bench press with *TrainerTap*

The bench press engages the pectoralis major (primary muscles) and the latissimus dorsi (secondary muscles) (Figure 2(b)). The system provides strong tapping at the peak contraction and relaxation points, helping users effectively activate their chest muscles while preventing excessive reliance on the triceps. Additionally, gentle

multiple taps follow the user's descending motion to ensure consistent tension is maintained throughout the exercise. For secondary muscles, strong single taps at both ends suffice, as the motion opposing gravity naturally maintains back muscle tension.

Deadlift with *TrainerTap*

The primary muscle in the deadlift is the latissimus dorsi, with the gluteal muscles as secondary (Figure 2©). Similar to the bench press, the system gives strong single taps at the lowest and highest points, emphasizing the user's cognitive involvement in the back movements. The gentle multiple taps start from the beginning to maintain tension in the back muscles, as the risk of injury is highest when the bar is at a range from ground to knee height. A strong single tap is also given to the gluteal muscles, which should be activated to support the back muscles initially.

4 CONCLUSIONS & FUTURE WORK

We propose *TrainerTap*, a weightlifting support system that simulates a personal trainer's tactile and auditory guidance to enhance the workout effects in solitary weightlifting. We developed a prototype and collaborated with trainers to design diverse tapping patterns for three popular workouts (squat, bench press, and deadlift). For future work, we will evaluate the system's efficacy and usability in solitary workouts as well as its applicability in trainer-led workouts. Furthermore, we will integrate the muscle sensor to realize the system's full potential in weightlifting through detailed and active feedback.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2022R1C1C1011962). This research was supported by the Chung-Ang University Research Grants in 2021.

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